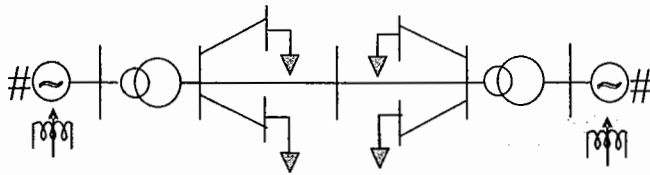


# 5

## Load flow studies

### Introduction

In a three phase ac power system active and reactive power flows from the generating station to the load through different networks buses and branches the flow of active and reactive power is called power flow (or) load flow.



The performance of a larger interconnected power system can be analyzed by evaluating the unknown electrical quantities.

Load flow study is the one, which can provide the solution of unknown electrical quantities, where the system is working at steady state condition along with certain inequality constraints.

#### (i) Voltage at the load point

$$V_{p \min} \leq V_p \leq V_{p \max}$$

$P$  is represented as bus

$$P = 1, 2, \dots, N$$

$n$  is total no of buses

#### (ii) Reactive power generated by generator

$$Q_{p \min} \leq Q_p \leq Q_{p \max}$$

#### (iii) Tap ratio of tap changing transformer

$$t_{p \min} \leq t_p \leq t_{p \max}$$

$$t \text{ is turn ratio} = \frac{1}{\text{turn ratio}}$$

The unknown electrical quantities can be obtained by solving so many no of network equations, which are developed either by using KCL or KVL because the network elements are represented in a lumped manner due to steady state condition.

The load flow study is the basis to make power system planning in order to meet the increased demand on the existing system.

#### (i) Addition of a new system to the existing system

#### (ii) Enhancing the capacity of existing system

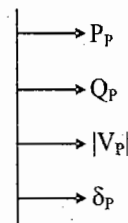
The load flow study is conducted by considering the balanced load on the existing system so that the power system is represented by a single line diagram and the network data is specified in a phase manner with an assumption that the system is a 3-phase star configuration.

In order to reduce the memory requirement to store the network data, it is preferred to develop the network equations by using KCL, where the network data is specified in the form of admittance connected to bus (OR) node.

The no. of network equations to be solved is more use any one of the interactive method to solve the network equations in order to reduce the time taking.

In steady state condition, there is a small and gradual variation of the load i.e.  $\Delta p_e$  and it can be ignored while solving the network equations of load flow steady

### Representation of Bus (OR) Node:



$P_p$  = Real power of Bus  $P$

$Q_p$  = Reactive power of Bus  $P$

$|V_p|$  = Magnitude of voltage at Bus  $P$

$\delta_p$  = phase angle of voltage at Bus  $P$

Out of the four electrical quantities, two of them are specified and two of them are unknown. The unknown electrical quantities can be obtained from known values along with admittance of the network.

### Classification of Busses:

1. Load bus: The real and reactive components can power are specified it is desired to fix out the voltage magnitude and phase angle through the load flow solution. It is required specify only  $P_D$  and  $Q_D$  such a bus as at a load bus voltage can be allowed to vary within permissible values.



- Generator bus (or) voltage controlled bus: The voltage magnitude corresponding to the generation voltage and real power  $P_G$  corresponding to its ratings is specified. It is required to fix out the reactive power generation  $Q_G$  and the phase angle of the bus voltages.
- Slack, swing (or) Reference bus: In power system there are mainly two types of buses load and generator buses. For these buses we have specified the real power  $P$  injections.

$$\text{Now } \sum_{i=1}^n P_i = \text{real power loss } P_L$$

Where  $P_i$  is the power injection at the buses

This is taken as positive for generator buses and negative for load buses.

| Bus Name                                                                                             | Specified values<br>(OR) known values | Unknown values         |
|------------------------------------------------------------------------------------------------------|---------------------------------------|------------------------|
| LOAD Bus<br>      | $P_P$ and $Q_P$                       | $ V_P $ and $\delta_P$ |
| Generator Bus<br> | $P_P$ and $ V_P $                     | $Q_P$ and $\delta_P$   |
| Slack (OR) Ref Bus                                                                                   | $ V_P $ and $\delta_P = 0$            | $P_P$ and $Q_P$        |
| Voltage controlled Bus                                                                               | $P_P$ and $ V_P $                     | $\delta_P, Q_P = 0$    |

Basically power system is having load buses and generator buses. Out of which 90% of the buses are load buses and 10% are generator buses.

#### Importance of slack bus:

Load flow studies are conducted in a balanced condition i.e. demand + loss = Generation

Losses are Transmission line losses

$$P_L = I^2 R$$

$$Q_L = \frac{1}{2} LI^2 - \frac{1}{2} CV^2$$

While solving the network equations of load flow study, the power at the bus is considered as

$$P_P = \sum P_{PG} - \sum P_{Pd} \quad \text{and}$$

$$Q_P = \sum Q_{PG} - \sum Q_{Pd}$$

Without consider the line losses under the assumption that the losses are less so that the load flow equations are simplified so that the time taking is less. However at the end of load flow study, the losses are considered by balancing the power equation.

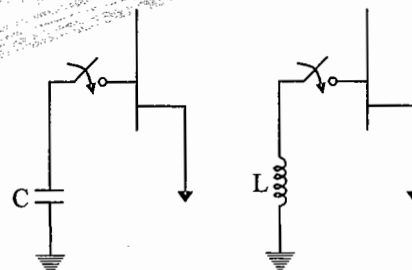
$$\sum P_{PG} - \sum P_{Pd} = P_L \quad \text{and}$$

$$\sum Q_{PG} - \sum Q_{Pd} = Q_L$$

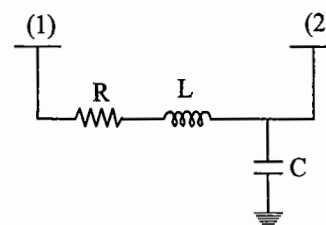
If the losses are considered, the generation will increase. In order to simplify the loss calculation the losses are supplied by one generator rather than distributing among all the generators. The generator which will supply the losses along with demand is called as slack bus. The angle of the slack bus i.e.  $\delta_P$  is assumed as zero, because it is the reference bus so that the angles of other buses are measured with respect to reference bus.

Generally the larger size of the generator is considered as slack bus because the percentage losses are less.

Voltage controlled bus is the load bus having shunt capacitor and shunt inductor, which can supply reactive power (OR) absorb reactive power so that the voltage is maintained as rated voltage. The net reactive power at the bus is zero. The angle of voltage magnitude is to be calculated

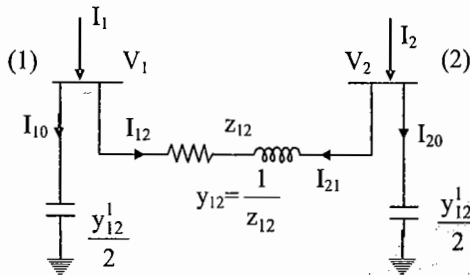


#### Node or bus admittance Network Equation





In load flow studies the shunt capacitance is replaced as half of the magnitude at each bus so that the transmission line is replaced by  $\pi$  network model and there is no change in no. of buses. When it is represented by T network model, the no. of buses will increase by one. So network model is not preferred.



$z_{12}$  is the impedance of an element connected between buses 1 and 2.

$$z_{12} = R + j\omega L = R + jx$$

$$y_{12} = \frac{1}{z_{12}} = G - j\beta$$

$y_{12}$  is the admittance of transmission line between buses 1 and 2.

In case of lossless transmission line

$$z_{12} = jx$$

$$y_{12} = -j\beta = y_{21} \text{ (bilateral network)}$$

$y_{12}^1$  is the line charge admittance due to shunt capacitance or line shunt susceptance.  $\frac{y_{12}^1}{2}$  is half line shunt susceptance.

At Node 1:

$$I_1 = I_{12} + I_{10}$$

$$I_1 = (V_1 - V_2)y_{12} + (V_1 - 0)\frac{y_{12}^1}{2}$$

$$I_1 = \left(y_{12} - \frac{y_{12}^1}{2}\right)V_1 - y_{12}V_2$$

$$I_{11} = Y_{11}V_1 + Y_{12}V_2$$

$$Y_{11} = \frac{y_{12}^1}{2} + y_{12}$$

It is called as total self admittance of bus1. It is nothing but sum of the admittances of the elements connected at bus 1. It has both series and shunt elements. It is having one element or more than one element. It is higher in magnitude.

$$Y_{12} = -y_{12} = Y_{21}$$

Total mutual admittance between buses 1 and 2 it is nothing but the admittance of the element with a negative sign. It has one element or zero element. It is having only series elements. It is having lesser in magnitude.

At Node (OR) Bus 2:

$$I_2 = I_{21} + I_{20}$$

$$I_2 = (V_2 - V_1)y_{21} + (V_2 - 0)y_{21}^1/2$$

$$I_2 = -y_{21}V_1 + (y_{21} + \frac{y_{21}^1}{2})V_2$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2 \text{ ----- (2)}$$

$$Y_{21} = -y_{21} = Y_{12}$$

$$Y_{22} = y_{21} + \frac{y_{21}^1}{2}$$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

$$I_{BUS} = Y_{BUS} V_{BUS}$$

$Y_{BUS}$  is bus admittance matrix

The current at Bus P

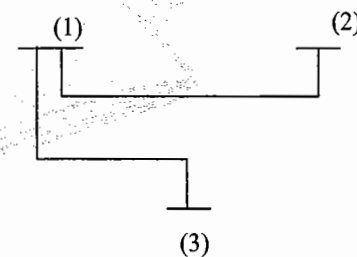
$$I_P = \sum_{q=1}^2 V_{pq} V_q$$

In case at n - bus system, the current at Bus P

$$I_P = \sum_{q=1}^n y_{pq} V_q$$

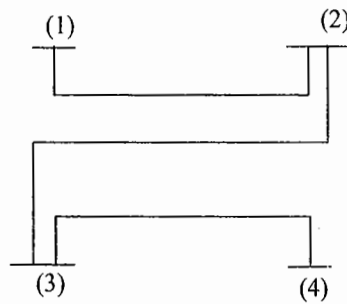
Bus (OR) Node admittance network equation

When the size of power system increases:



$$\begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} & Y_{13} \\ Y_{21} & Y_{22} & 0 \\ Y_{31} & 0 & Y_{33} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix}$$

It is observed that some of the mutual admittances are zero, but total self admittance are non zeros



$$\begin{bmatrix} I_1 \\ I_2 \\ I_2 \\ I_4 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} & 0 & 0 \\ Y_{21} & Y_{22} & 0 & Y_{24} \\ 0 & 0 & Y_{33} & Y_{34} \\ 0 & Y_{42} & Y_{43} & Y_{44} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix}$$

The mutual elements which are become zero and increased non zero mutual elements are reduced. In a larger PS network 90% and above are become zero and the remaining are non zeros. When the matrix is having more no of elements as zeros such a matrix is called sparse matrix. The  $Y_{BUS}$  matrix is also called sparse matrix and the memory recurrent to store  $Y_{BUS}$  elements is less. Hence the load flow studies are conducted by using KCL technique.

The electrical network is a bilateral network so that  $Y_{BUS}$  matrix is also symmetric matrix and the memory requirement is further reduced because it is sufficient to store either upper triangle (OR) lower triangle elements only.

When there are no shunt elements at the buses, the sum of the elements in the row are zero. The determinant of the matrix is zero such a matrix is called singular matrix.

$$Y_{11} = y_{12} + \frac{y_{12}^1}{2}$$

$$Y_{12} = -y_{12}$$

No shunt element that is  $\frac{y_{12}^1}{2}$  is not present

$$Y_{11} = y_{12}$$

$$Y_{11} + Y_{12} = 0$$

Modifications of  $Y_{BUS}$ : When a transmission line is removed between any two buses, the following changes are taking place

- Mutual admittances of the two buses are become zero
- Total self admittances of the two buses are reduced in the magnitude

The size of  $Y_{BUS}$  is same and there are no changes in other elements of  $Y_{BUS}$

Ex: Transmission line is removed between buses 1 and 2

$$Y_{12} = Y_{21} = 0$$

$$Y_{11 \text{ mod}} = Y_{11 \text{ old}} - y_{12}$$

$$Y_{22 \text{ mod}} = Y_{22 \text{ old}} - y_{12}$$

#### Addition of external shunt elements at the buses:

When the external shunt element is added at any one of the bus, the total self admittance of the bus is modified

Ex: The external shunt inductor is added at bus 1

$$Y_{11 \text{ mod}} = Y_{11 \text{ old}} + y_{01}$$

$y_{01}$  is the admittance of external shunt inductor

Addition of another transmission line between two buses

When the transmission line is added between any two buses, the following modifications are taken place.

$$Y_{11 \text{ mod}} = Y_{11 \text{ old}} + y_{12(z)}$$

$$Y_{22 \text{ mod}} = Y_{22 \text{ old}} + y_{21(2)}$$

$$Y_{12 \text{ mod}} = Y_{12 \text{ old}} = Y_{12 \text{ old}} - y_{12(2)}$$

#### Example 5.1:

The power systems network of dealing with 20 bus. The  $Y_{bus}$  matrix of the system is 80 % Sparse. the minimum number of transmission line that are existing in the bus power system.

$$\text{Sol: } Y_{BUS} = 20 \times 20 = 400$$

$$\text{Non-zero} = 0.2 \times 400 = 80$$

$$\text{Diagonal} = 20$$

$$\text{Mutual} = 60 \text{ (non-zero)}$$

$$\text{Lines} = \frac{\text{mutual}}{2} = \frac{60}{2} = 30$$

#### Example 5.2.

The power system network if having a 100 bus system. The  $Y_{bus}$  of the power system network if having 90 % sparse. The minimum number transmission lines that are exist in the bus system.

$$\text{Sol: } Y_{bus} = 100 \times 100 = 10^4$$

$$\text{Non zero} = 0.1 \times 10^4 = 10^3$$

$$\text{Diagonal} = 100$$

$$\text{Mutual} = 900$$

$$\text{Lines} = \frac{\text{mutual}}{2} = \frac{900}{2} = 450$$



### Construction of Bus Impedance Matrix:

The purpose of bus admittance matrix is for load flow studies in a steady state condition, and Bus impedance matrix is for the calculation of short circuit currents during the short circuit fault. The Bus impedance matrix can be obtained by using Bus admittance matrix as

$$Z_{bus} = Y_{bus}^{-1}$$

The above conversion is not accurate, because the effect of shunt capacitance of transmission line and any external shunt elements at the buses are considered during the load flow studies, but they are ignored during the short circuit studies. The construction of Bus impedance matrix can be dealt separately.

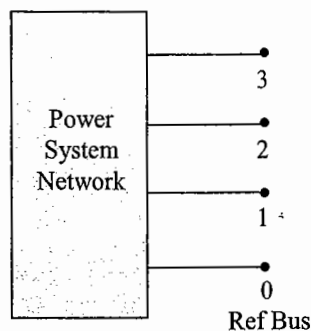
### Procedure for the construction of $Z_{Bus}$ :

While constructing Bus impedance, addition of

One element is taken place at any time between two busses and complete the addition of all elements in the given bus system. because the short circuit fault at one bus will also effect the voltages at the other busses. In case of load flow studies, the load at one bus does not effect the voltages at other busses. Hence all the buses are considered at the same time in case of bus admittance matrix.

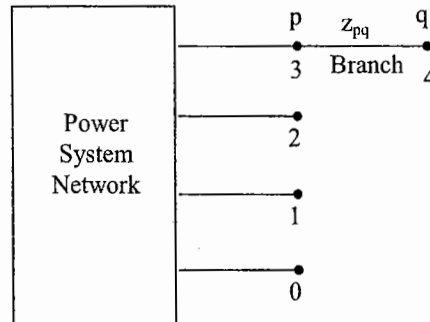
Case (i):

If the added element from any one of the bus does create a new bus to the existing system, it is treated as branch. The size of  $Z_{Bus}$  is increased by one. The impedance of new bus are to be calculated without changing the original  $Z_{Bus}$ .



$$Z_{Bus} = \begin{bmatrix} Z_{11} & Z_{12} & Z_{13} \\ Z_{21} & Z_{22} & Z_{23} \\ Z_{31} & Z_{32} & Z_{33} \end{bmatrix}$$

One of the bus is taken as reference bus, while constructing  $Z_{Bus}$  and it is zero potential bus i.e. ground. It is normally a shunt element i.e. generator impedance.



$$Z_{Bus} = \begin{bmatrix} Z_{11} & Z_{12} & Z_{13} & Z_{14} \\ Z_{21} & Z_{22} & Z_{23} & Z_{24} \\ Z_{31} & Z_{32} & Z_{33} & Z_{34} \\ Z_{41} & Z_{42} & Z_{43} & Z_{44} \end{bmatrix}$$

$Z_{41}, Z_{42}$  and  $Z_{43}$  are mutual impedances and  $Z_{44}$  is self impedance of the bus.

In a general case the busses are considered as p and q.

### Mutual Impedances:

$$Z_{qi} = Z_{pi} \\ \text{Where } i = 1, 2, \dots, n \\ \neq q$$

### Self Impedances:

$$Z_{qq} = Z_{pq} + z_{pq}$$

$z_{pq}$  is the impedance of added branch element

The above equations are applicable if the bus P is not a reference bus i.e. zero bus.

In case, Bus P is a reference bus

### Mutual Impedances:

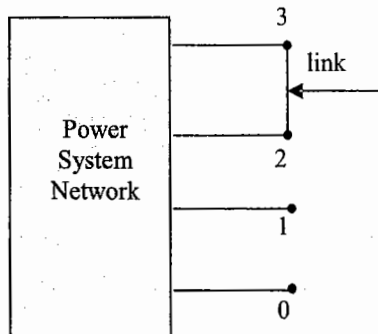
$$Z_{qi} = 0, Z_{pi} = 0$$

### Self Impedances:

$$Z_{qq} = z_{pq}, Z_{pq} = 0$$

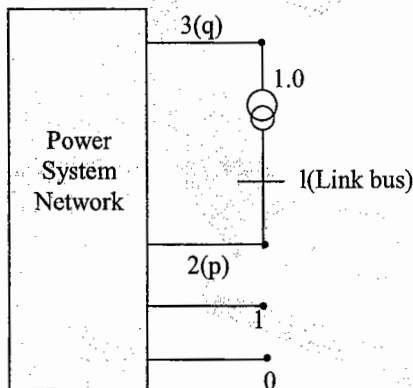
Case (ii):

If the added element from any one of the bus does not create a new bus to the existing bus system, it is treated as link and the size of  $Z_{Bus}$  will remain same. All the elements of original  $Z_{Bus}$  are to be modified to consider the effect of the impedance of added link element.



#### Procedure for modification of $Z_{Bus}$ :

Connect a voltage source in series with the added element having the magnitude of 1.0 P.U. So that a temporary bus is created, which is called as link bus. The size of  $Z_{Bus}$  is increased temporarily by one. The impedances of the temporary are to be calculated.



$$Z_{Bus} = \begin{bmatrix} Z_{11} & Z_{12} & Z_{13} & Z_{1l} \\ Z_{21} & Z_{22} & Z_{23} & Z_{2l} \\ Z_{31} & Z_{32} & Z_{33} & Z_{3l} \\ Z_{l1} & Z_{l2} & Z_{l3} & Z_{ll} \end{bmatrix}$$

#### Mutual impedance:

$$Z_{li} = Z_{pi} - Z_{qi}$$

$$i = 1, 2, \dots, n$$

$$\neq l$$

#### Self Impedance:

$$Z_{ll} = Z_{pl} - Z_{ql} + Z_{pl}$$

$Z_{pl}$  is impedance of added link element. The above equations are correct if bus 'p' is not a reference bus.

In case bus 'p' is a reference bus

#### Mutual impedance:

$$Z_{li} = -Z_{qi}, Z_{pi} = 0$$

#### Self impedance:

$$Z_{ll} = -Z_{ql} + Z_{pl}, Z_{pl} = 0$$

The original size  $Z_{bus}$  can be obtained by eliminating the link bus for which the voltage source is short circuited.

$$Z_{Bus \text{ mod}} = Z_{Bus \text{ old}} - \frac{[Z_{li}][Z_{lj}]}{Z_{ll}}$$

#### Case 3:

Removal of transmission line between any two buses.

If the transmission line is removed between any two buses, the size of  $Z_{bus}$  can remain same i.e. the lines which are permitted to remove are link elements. When the link element is removed, the entire  $Z_{bus}$  is to be modified. The procedure for modification of  $Z_{bus}$  is same as the procedure for addition of link element.

#### Mutual Impedance:

$$Z_{li} = Z_{pi} - Z_{qli}$$

#### Self Impedance:

$$Z_{ll} = Z_{pl} - Z_{qli} - Z_{pl}$$

$Z_{pl}$  is the impedance of removed link element.

$$Z_{Bus \text{ mod}} = Z_{Bus \text{ old}} - \frac{[Z_{li}][Z_{li}]}{Z_{ll}}$$

#### Example 5.3:

The  $Z_{Bus}$  of a system is

$$Z_{Bus} = \begin{bmatrix} 0.1 & 0.1 & 0.1 \\ 0.1 & 0.2 & 0.1 \\ 0.1 & 0.1 & 0.3 \end{bmatrix} \text{ p.u.}$$

If a 3-phase fault occurs at Bus-2, the p.u fault current in each phase is

$$\text{Sol: } I_f = I_R = I_{R1} = \frac{E_{R1}}{X_{1eq}} = \frac{E_{R1}}{z_{22}} = \frac{1.0}{0.2} = 5.0$$



**Example 5.4:**

The positive sequence bus impedance matrix of a 4-Bus power system network is given below:

$$Z_{1, Bus} = \begin{bmatrix} j0.724 & j0.620 & j0.656 & j0.644 \\ j0.620 & j0.738 & j0.642 & j0.660 \\ j0.656 & j0.642 & j0.0702 & j0.676 \\ j0.664 & j0.660 & j0.676 & j0.719 \end{bmatrix}$$

If a symmetrical fault occurs on Bus-2, then positive sequence component of fault current is

$$\text{Sol: } I_f = I_R = I_{R1} = \frac{E_{R1}}{Z_{22}} = \frac{1.0}{j0.738} = -j1.35$$

| $Y_{BUS}$                                           | $Z_{BUS}$                                         |
|-----------------------------------------------------|---------------------------------------------------|
| (i) Memory requirement to store the element is less | (i) More memory requirement to store two elements |
| (ii) It is useful for load flow studies             | (ii) It is useful short circuit studies           |
| (iii) Shunt element are included                    | (iii) Shunt elements are ignored                  |

| Gauss-seidal method                                                                                                                     | Newton Raphson method                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (i) It gives the solution of non linear simultaneous equation in rectangular coordinates.                                               | (i) It gives the solution of non linear simultaneous equations in rectangular (OR) polar form                                                               |
| (ii) Convergent criteria is a quadratic<br>$\Delta V_p =  V_p^k  -  V_p^{k-1} $                                                         | (ii) Convergent criteria is a quadratic<br>$\Delta P_p = P_{p \text{ spec}} - P_{p \text{ cal}}$ (OR) $\Delta Q_p = Q_{p \text{ spec}} - Q_{p \text{ cal}}$ |
| (iii) It is easy in programming load flow equations so that the memory requirement is less to store the data                            | (iii) It is difficult in programming load flow equations so that the memory requirement is high.                                                            |
| (iv) The time taken for each iteration is less                                                                                          | (iv) Time taken for each iteration is high                                                                                                                  |
| (v) No of iteration required are more to get the convergent criteria and the no of iterations does depends on the size of power system. | (v) No of iterations required are less to get the convergent criteria and no of iteration are not depends on size of power system                           |
| (vi) Total time taken to get the convergent criteria is high                                                                            | (vi) Total time taken to get the convergent criteria is less                                                                                                |
| (vii) Selection slack will affect the convergent criteria                                                                               | (vii) Selection slack Bus will not affect the convergent criteria                                                                                           |
| (viii) Un reliable convergent                                                                                                           | (viii) Reliable convergent                                                                                                                                  |
| (ix) Applicable for small power systems                                                                                                 | (ix) Applicable to larger power systems                                                                                                                     |